



**İstanbul
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DIAGNOSING THE BREATHING SOUNDS AS COPD OR ASTHMA

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1 Realistic Constrains

1.1 Social, Environmental and Economical Impact

For an engineering project to be successful, it has to have beneficial effects to public in a shape or form. In recent years, deep learning became more accessible and reliable in various fields. From healthcare to economy, from image recognition to natural language processing, from recommendation systems to speech recognition, it found a lot of topics to tackle.

Tying into the medical field with deep learning, we can analyze the patients' symptoms, tests, and give results or feedback based on the inputs beforehand. Early detection on the diseases could save lives and let the patients to get the needed care in short time. With that in mind, aim of the project was tackling detection of the various diseases depending on sound recordings. Also, reducing the costs on the diagnosis detections. By this way, diagnosis detection techniques use the advantage of automated systems and build a more efficient future.

1.2 Cost and Time Analysis

In order to estimate the project's cost, the problem was splitted into a few steps. Deep learning heavily relies on the electrical power. However, since using the free version of Google Collab, which is a cloud service, the training models are not trained in the local environment of our computers. The service itself has a subscription option, which allows for faster GPUs and more RAM. If the project were expanded, there would be need to have either getting a designated PC for the model training or continue using Collab's upper tier services, which starts from 138₺/per month. If the subscription option were picked, with monthly fee, it would sum up to pay 1,104₺, beginning from start of the project. If we abandoned the cloud option and decide to get personal computers for two people that are doing the project, We would need up to date model GPUs and processors. Without diving deep down, a latest gen AMD Ryzen or Intel processor such as i9 or i7, would cost us around 9000₺, and a NVIDIA Geforce 3090 graphics card would cost at least 34000₺, which without adding other components, a single PC would sum up to at least 43000₺. A cheaper option that that would be getting a prebuilt PC, or if a budget is considered, with making concessions on model training time, a laptop could be an option, with a different generation CPU and processor. If we did a time analysis on our labor, an average hourly income would be 33₺. When we multiply that value with our labor, it is 3300₺ per person, and sums up to 6600₺ for total. Whole the project application process took

5 months approximately, except the theoretical researches part.

1.3 Standards

In this section, the standards of NSPE Code Of Ethics and IEEE Standards for Software and System Engineering are held.

1.3.1 NSPE Code Of Ethics

Code of ethics are ;

- Engineers shall avoid deceptive acts.
- Engineers should act responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession. [1]

As professional obligations:

- Engineers shall be guided in all their relations by the highest standards of honesty and integrity. [1]
 - a. Engineers shall acknowledge their errors and shall not distort or alter the facts.
- Engineers shall avoid all conduct or practice that deceives the public.

1.3.2 IEEE Standards

- IEEE/ISO/IEC 16085-2020 - ISO/IEC/IEEE International Standard - Systems and software engineering – Life cycle processes – Risk management[2]
- IEEE 15026-2-2011 - IEEE Standard–Adoption of ISO/IEC 15026-2:2011 Systems and Software Engineering–Systems and Software Assurance–Part 2: Assurance Case [2]
 - This document is suitable for the management of all risks encountered in any organization or project appropriate to the systems or software projects regardless of context, type of industry, technologies utilized, or organizational structures involved. [3]

2 Risk Analysis And Changes

In this section, We can analyze types of risks in some parts:

- Risk 1: Amount of recordings that needed to train the models.

Change: Regarding this risk, we used sound augmentations to create inputs for our mode.

- Risk 2: Misperforming of the model, therefore wrong prediction.

Change: Different neural networks were used, and the results were double checked.

- Risk 3: If a wrong prediction occurs related to any misperforming, a wrong diagnosis and treatment way might be chosen. So, the doctor's extra diagnosis might be required in this situation.

Change: The model performance was tried to be maximized to reduce maximal misdiagnosis probability.

2.1 Challenges

In this section we can summarize the challenges we faced:

- Fairness: Deep learning models are complicated and should stay clear of biased outputs. It can harm results, therefore in our case, it would mislabel a disease and it would lead to mistreatment of the sickness.
- Quality of the data: If the said data has faults, errors, or in our case, different sounds, our results would be compromised and mislabeled.
- Quantity of the data: Deep learning models need a lot of examples to function properly.
- Underfitting: Occurs when the model is not making good progress on both testing and learning data set.
- Overfitting: Occurs when the model is not making high accuracy predictions on testing data.

2.2 Weak And Strong Points

2.2.1 Weak Points

- Unavailability of medical data sets: In general, medical field tends to have less data from other fields. Usually there is no shared data set

on topics, such as ours. We just used publicly available sound data set and no other.

- Size of the data set: Compared to an image processing or most of the sound recordings data sets, ours is relatively small. Some diseases has low amount of sound recordings or vice versa. For example, asthma had only 1 recording and COPD had nearly 800 recordings.

2.2.2 Strong Points

- This study is easily accessible,
- Respiratory Sound Database[4] is available publicly and every sound recording has its own diagnosed disease labeled.
- The model results were highly accurate.

3 Contributions

- Author 1 Burakcan Türkan → mainly worked on the practical applications(code) of several models.
- Author 2 (Ahmet Gökay Ateş) → mainly worked on the research of necessary topics, theoretical search, detailing and writing the project report with related works.
- Supervisor 1 (Elena Battini Sönmez) → Supervised with the weekly meetings, did the consultation and gave many feedbacks.
- Supervisor 2 (Özgür Özdemir) → Supervised with the weekly meetings, helped with the theoretical and applicational parts of the study.

4 Conclusion

Deep learning is an inevitable method in today's world. It runs on many areas. By using it, building a cheaper, faster, and stronger future is possible. From the beginning of this study, publicly free and available tools were used. If the sources hadn't been used in that way, this project's cost would have been a little bit expensive. Issues with the availability of model training was overcome by using Google Colab. The engineering codes of ethics and standards were followed in whole parts of the study. There were some possible risky points in the study like model insufficiency. However, that risks were

already tried to be minimized. While aiming to get better results, many obstacles were faced and challenged. The problem of having a small sample size was solved with some sound augmentation techniques. Some kinds of model problems were handled such as overfitting or underfitting. After all the challenges, the model success rate was taken to the maximum efficiency point in the current circumstances. The whole study is easily observable, the used data set is a public one related to the hardness of accessing the private medical data sets. However, current best model results are satisfying. This project was supervised by two instructors and developed and reported by two authors.

References

- [1] NSPE.org, “Nspe code of ethics for engineers.” <https://www.nspe.org/resources/ethics/code-ethics>, Last accessed 30 June 2021.
- [2] I. S. Association, “Iso/iec/ieee international standard - systems and software engineering – life cycle processes – risk management, last accessed 30 june 2021.” <https://standards.ieee.org/ieee/16085/6932/>, Published 15/01/2021.
- [3] “Iso/iec/ieee international draft standard - systems and software engineering - life cycle processes - risk management,” *ISO/IEC/IEEE P16085/CD3*, August 2019, pp. 1–68, 2019.
- [4] B. Rocha, D. Filos, L. Mendes, G. Serbes, S. Ulukaya, Y. Kahya, N. Jakovljevic, T. Turukalo, I. Vogiatzis, E. Perantoni, E. Kaimakamis, P. Natsiavas, A. Oliveira, C. Jácome, A. Marques, N. Maglaveras, R. P. Paiva, I. Chouvarda, and P. de Carvalho, “An open access database for the evaluation of respiratory sound classification algorithms,” *Physiol Meas*, vol. 40, 2019.